# OPENNESS OF VERSALITY FROM A PERFECT DUAL OBSTRUCTION THEORY

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 4714-4722 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 4724-4749, the complete author proof (24 body lines).

*Editorial note.* Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 31 of queue_artin.json (pb=1322, stmt 4714-4722, proof 4724-4749).

*Editorial note.* stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 07YY: sources/stacks-artin/pinned-original-source.tex, source lines 4616–4628. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 07YN: sources/stacks-artin/pinned-original-source.tex, source lines 4364–4381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 07VI: sources/stacks-artin/pinned-original-source.tex, source lines 4749–4749. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 07YX: sources/stacks-artin/pinned-original-source.tex, source lines 4570–4585. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *In Situation 07YX assume that (iv) of Lemma 07YY holds and that $K^\bullet$ is a perfect object of $D(A)$. In this case, if $x$ is versal at a closed point $u_0 \in U$ then there exists an open neighbourhood $u_0 \in U' \subset U$ such that $x$ is versal at every finite type point of $U'$.*

*Proof.* We may assume that $K^\bullet$ is a finite complex of finite projective $A$-modules. Thus the derived tensor product with $K^\bullet$ is the same as simply tensoring with $K^\bullet$. Let $E^\bullet$ be the dual perfect complex to $K^\bullet$, see More on Algebra, Lemma 07VI. (So $E^n = \mathrm{Hom}_A(K^{-n}, A)$ with differentials the transpose of the differentials of $K^\bullet$.) Let $E \in D^-(A)$ denote the object represented by the complex $E^\bullet[-1]$. Let $\xi \in H^1(\mathrm{Tot}(K^\bullet \otimes_A NL_{A/\Lambda}))$ be the element constructed in Lemma 07YY and denote $\xi : E = E^\bullet[-1] \to NL_{A/\Lambda}$ the corresponding map (loc.cit.). We claim that the pair $(E, \xi)$ satisfies all the assumptions of Lemma 07YN which finishes the proof.

Namely, assumption (i) of Lemma 07YN follows from conclusion (1) of Lemma 07YY and the fact that $H^2(K^\bullet \otimes^{\mathbf{L}}_A -) = \mathrm{Ext}^1(E, -)$ by loc.cit. Assumption (ii) of Lemma 07YN follows from conclusion (2) of Lemma 07YY and the fact that $H^1(K^\bullet \otimes^{\mathbf{L}}_A -) = \mathrm{Ext}^0(E, -)$ by loc.cit. Assumption (iii) of Lemma 07YN is clear. □